# MCA 211 : Environmental Science & Ethics

**Unit-I :**
*Knowing our environment, socio-economic, political & environmental situations of India,* Population, The land, Forest & wild life, energy sources, Toxic Chemicals & Toxicity, Hazardous waste, sustainability and Human Development.

**Unit-II**
Ecology, Principles of Energy and Matter in the environment, Food Chain, Carbon and Nitrogen cycles, Biodiversity, benefits and what threatens it, Causes of extinction of the species and their remediation.

**Unit-III**
Perspectives of global environmental issues, the green house effect, sources of carbon emissions, chlorofluorocarbons, their major uses and combined effect of green house gases, stratosphere, Ozone depletion and impacts.

**Unit-IV**
Environmental Pollution; causes, pollutants their effect and control of pollution – (i) Air (ii) Water (iii) Soil (iv) Noise (v) Thermal (vi) Marine, the legislation of air & water pollution, solid and/ hazardous waste disposal.

**Unit-V**
Social issues and environment, the western perspective, Environmental ethics, conservation of natural resources, Effectiveness of various religions in environmental conservation, technological progress and quality of life, harmony between materialism and spiritualism, imbibing human values in environmental studies, NGOs and their role in environmental preservations.

**Text Books**


1. Environmental Studies – Benny Joseph – Tata McgrawHill-2005
2. Environmental Studies – Dr. D.L. Manjunath, Pearson Education-2006.
3. Environmental studies – R. Rajagopalan – Oxford Publication - 2005.
4. Text book of Environmental Science & Technology – M. Anji Reddy – BS Publication.


**References Books :**


1. Principles of Environmental Science, Curnningham. W.P. & Cunningham M.A., Tata McGraw Hill Publishing Co. Ltd., New Delhi.
2. India's Environmental Policies, Programmes & Stewardship by Prof. O.P. Dwivedi, Macmilan Press Ltd., London.
3. Textbook of Environment & Ecology by Deeksha Dave and S.S. Katewa, Cengage Learning India Pvt. Ltd., Patparganj, Delhi, 2009.
4. Principles of Environmental Science and Engineering – P. Venugoplan Rao, Prentice Hall of India.
5. Environmental Science & Engineering – Meenakshi, Prentice Hall India.